

Microsoft Entra IAM & PAM Security Lab

Identity & Access Management • Privileged Identity Management

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Overview

This project documents a practical Microsoft Entra ID identity-security lab covering identity and access management (IAM), role-based access control (RBAC), least privilege, multi-factor authentication (MFA), authentication methods, sign-in investigation, audit-log investigation, Privileged Identity Management (PIM) evaluation, and PAM/JIT access concepts.

The lab was built in a Microsoft Entra ID Free tenant using separate test identities to simulate an organization's security environment.

Lab Scenario

Objective: build a small identity-security environment and investigate how privileged access can be controlled and monitored.

Test Identities

Identity	Purpose
Rahul Singh	Security analyst / privileged-access test user
Niha Sharma	Standard employee test user
Ishaan Malhotra	Tenant administrator / lab administrator

A dedicated set of security groups (GG-Employees, GG-IT-Admins, GG-Security-Team) was also created for the lab.

Lab Environment Setup

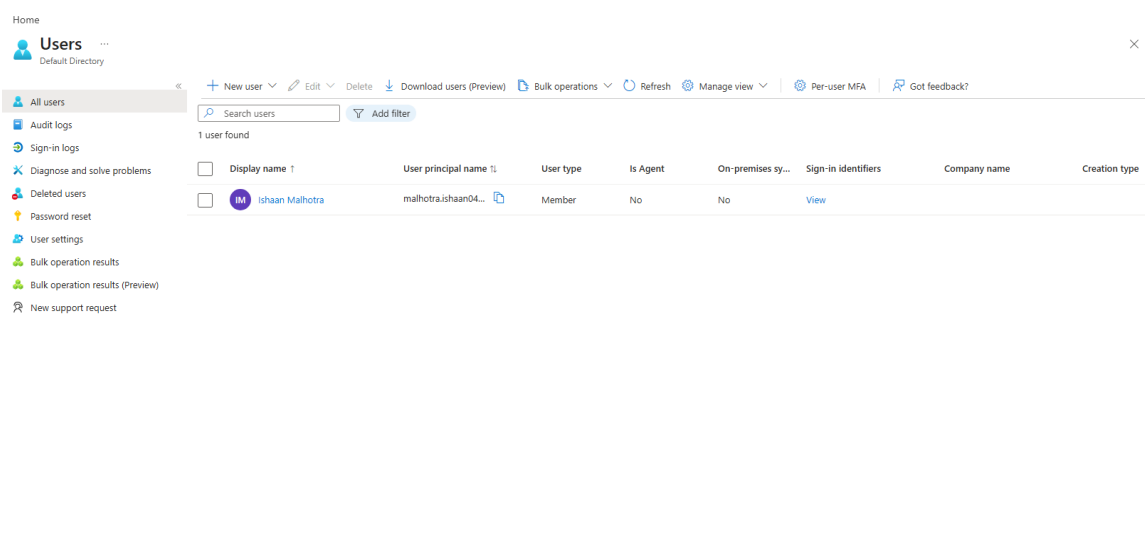


Figure 1. Entra tenant user list — initial administrator identity (Ishaan Malhotra).

Create new user ...

Create a new internal user in your organization

Basics Properties Assignments **Review + create**

Basics

User principal name	employee01@malhotraishaan04gmail.onmicrosoft.com
Display name	Niha Sharma
Mail nickname	employee01
Password	
Account enabled	Yes

Properties

User type	Member
-----------	--------

Assignments

Administrative units

Groups

Roles

Figure 2. Creating the standard employee test identity, Niha Sharma.

Niha Sharma

User

Search

Edit propertiesDeleteRefreshReset passwordRevoke sessionsManage viewGot feedback?

Overview

Audit logs

Sign-in logs

Diagnose and solve problems

Custom security attributes

Assigned roles

Administrative units

Groups

Applications

Licenses

Devices

Azure role assignments

Authentication methods

New support request

OverviewMonitoringProperties

Basic info

NS

Niha Sharma

employee01@malhotraishaan04gmail.onmicrosoft.com

Member

User principal name	employee01@malhotraishaan04gmail.onmicrosoft.com	Group memberships	1
Object ID	f4274470-fd98-4a4d-80ce-2e7f5390121a	Applications	0
Created date time	25 Aug 2026, 23:47	Assigned roles	0
User type	Member	Assigned licenses	0
Sign-in identifiers	View		

My Feed

Account status

Enabled

Edit

B2B invitation

Convert to external user

Figure 3. Niha Sharma's user profile after account creation.

New groupDownload groupsRefreshManage viewDeleteGot feedback?

Search

Add filter

Search modeContains

3 groups found

☐	Name ↑	Object id	Group type	Membership type	Email
☐	GG-Employees	2f7c0c55-e572-4b62-8778-23b03c20d5a9	Security	Assigned	
☐	GG-IT-Admins	55fd55d8-c311-41a3-bd16-b45d09e9917c	Security	Assigned	
☐	GG-Security-Team	7a1a2159-e4be-41e9-9f32-941c7d81a7d4	Security	Assigned	

Figure 4. Security groups created for the lab: GG-Employees, GG-IT-Admins, GG-Security-Team.

GG-Employees | Members

Group

« + Add membersBulk operationsRefreshManage viewRemoveGot feedback?

Overview

Diagnose and solve problems

Manage

Properties

Members

Owners

Roles and administrators

Administrative units

Group memberships

Applications

Licenses

Azure role assignments

Activity

Privileged Identity Management

Access reviews

Audit logs

Bulk operation results

Troubleshooting + Support

New support request

Direct membersAll members

Search

Add filter

1 group member found

☐	Name ↑	Type	Email	User type	Object id
☐	Niha Sharma	User		Member	f4274470-fd98-4a4d-80ce-2e7f5390121a

Figure 5. GG-Employees group membership showing Niha Sharma.

GG-Security-Team | Members

Group

« + Add membersBulk operationsRefreshManage viewRemoveGot feedback?

Overview

Diagnose and solve problems

Manage

Properties

Members

Owners

Roles and administrators

Administrative units

Group memberships

Applications

Licenses

Azure role assignments

Activity

Privileged Identity Management

Access reviews

Audit logs

Bulk operation results

Troubleshooting + Support

New support request

Direct membersAll members

Search

Add filter

1 group member found

☐	Name ↑	Type	Email	User type	Object id
☐	Rahul Singh	User		Member	70c38c66-a5ff-474b-9182-ae479d7fe83b

Figure 6. GG-Security-Team group membership showing Rahul Singh.

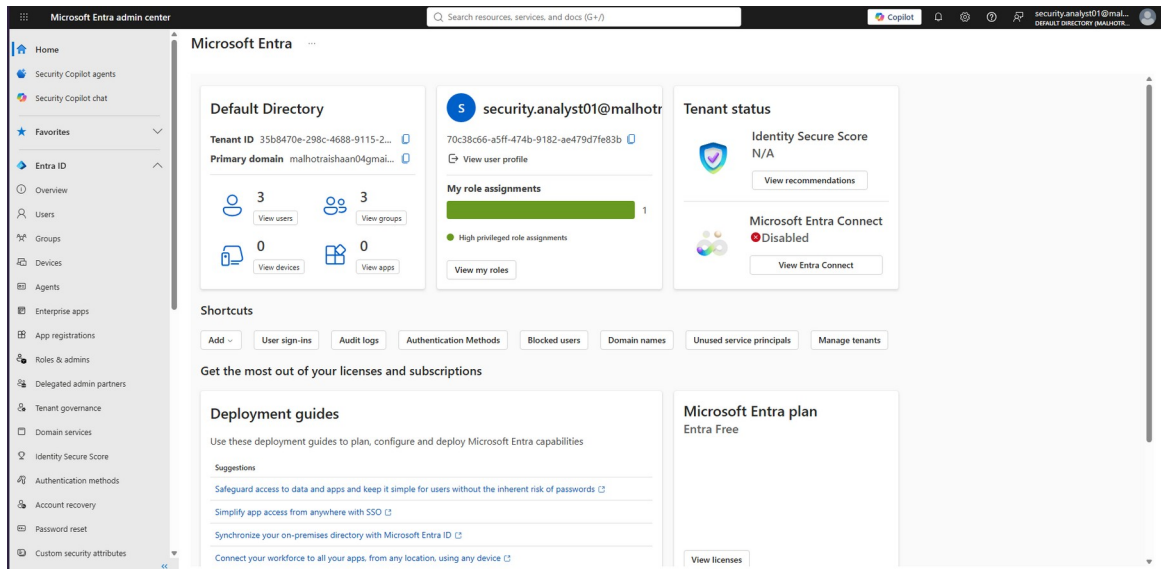


Figure 7. Tenant overview dashboard, viewed from Rahul Singh's session.

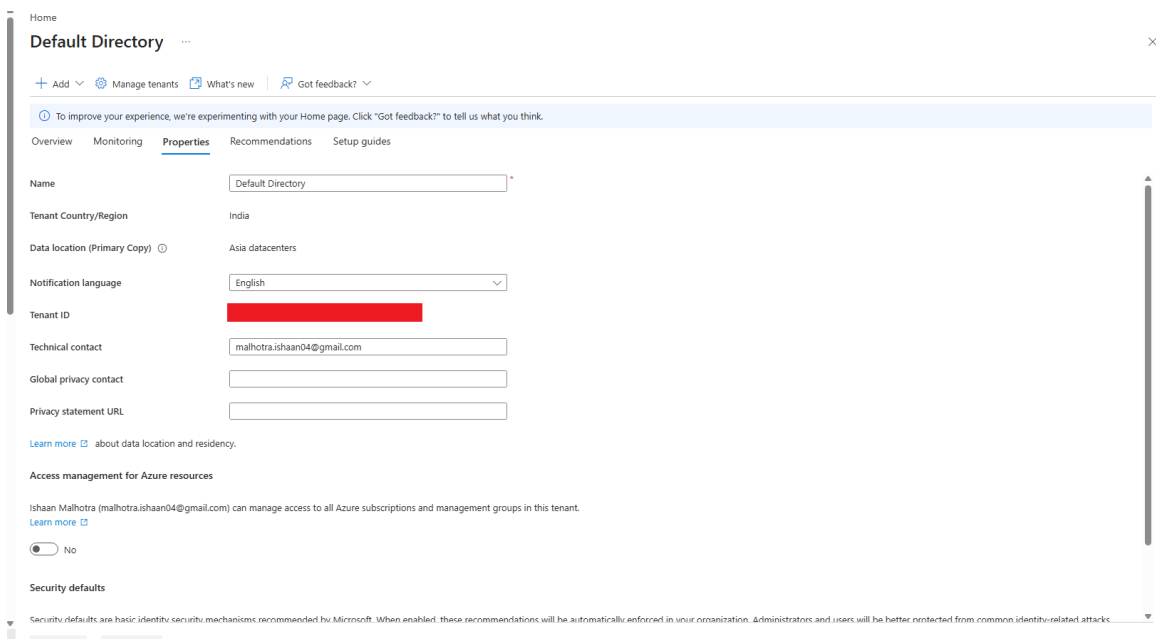


Figure 8. Tenant properties confirming the Microsoft Entra ID Free licensing plan (Tenant ID redacted).

1. IAM Implementation — RBAC

Role-Based Access Control

A Security Reader role was assigned to Rahul Singh. Security Reader provides read access to security information and reports without granting broad administrative control.

Roles and administrators | All roles

To create custom roles, your organization needs Microsoft Entra ID Premium P1 or P2. Start a free trial. →

Your Role: Global Administrator and 1 other roles

Administrative roles
Administrative roles are used for granting access for privileged actions in Microsoft Entra ID. We recommend using these built-in roles for delegating access to manage broad application configuration permissions without granting access to manage other parts of Microsoft Entra ID not related to application configuration. [Learn more](#).

[Learn more about Microsoft Entra ID role-based access control](#)

Search by name or description Add filters

Role	Description	Privileged	Ass...	Type
☐ Agent ID Administrator	Manage all aspects of agents in a tenant including identity lifecycle operations for agent blueprints, agent service principals, agent identities, and agent users.	PRIVILEGED	0	Built-in
☐ Agent ID Developer	Create an agent blueprint and its service principal in a tenant. User will be added as an owner of the agent blueprint and its service principal.		0	Built-in
☐ Agent Registry Administrator	Manage all aspects of the Agent Registry service in Microsoft Entra ID		0	Built-in
☐ AI Administrator	Manage all aspects of Microsoft 365 Copilot and AI-related enterprise services in Microsoft 365.	PRIVILEGED	0	Built-in
☐ AI Reader	Read all aspects of Microsoft 365 Copilot and AI-related enterprise services in Microsoft 365.	PRIVILEGED	0	Built-in
☐ Application Administrator	Can create and manage all aspects of app registrations and enterprise apps.	PRIVILEGED	0	Built-in
☐ Application Developer	Can create application registrations independent of the 'Users can register applications' setting.	PRIVILEGED	0	Built-in
☐ Attack Payload Author	Can create attack payloads that an administrator can initiate later.		0	Built-in
☐ Attack Simulation Administrator	Can create and manage all aspects of attack simulation campaigns.		0	Built-in
☐ Attribute Assignment Administrator	Assign custom security attribute keys and values to supported Microsoft Entra objects.		0	Built-in
☐ Attribute Assignment Reader	Read custom security attribute keys and values for supported Microsoft Entra objects.		0	Built-in
☐ Attribute Definition Administrator	Define and manage the definition of custom security attributes.		0	Built-in
☐ Attribute Definition Reader	Read the definition of custom security attributes.		0	Built-in
☐ Attribute Log Administrator	Read audit logs and configure diagnostic settings for events related to custom security attributes.		0	Built-in
☐ Attribute Log Reader	Read audit logs related to custom security attributes.		0	Built-in
☐ ...	Read and edit the provisioning configuration of all active custom security attributes for an			

Figure 9. Full built-in role list, viewed as Global Administrator.

Security Reader | Assignments

You can also assign built-in roles to groups now. [Learn More](#)

Successfully added assignment
Successfully added assignment Rahul Singh

Search by name

Type

All

Name	Username	Type	Scope
☐ Rahul Singh	security.analyst01@malhotraishaan04gmail.onmicrosoft.com	User	Directory

Figure 10. Security Reader role assignment successfully added for Rahul Singh.

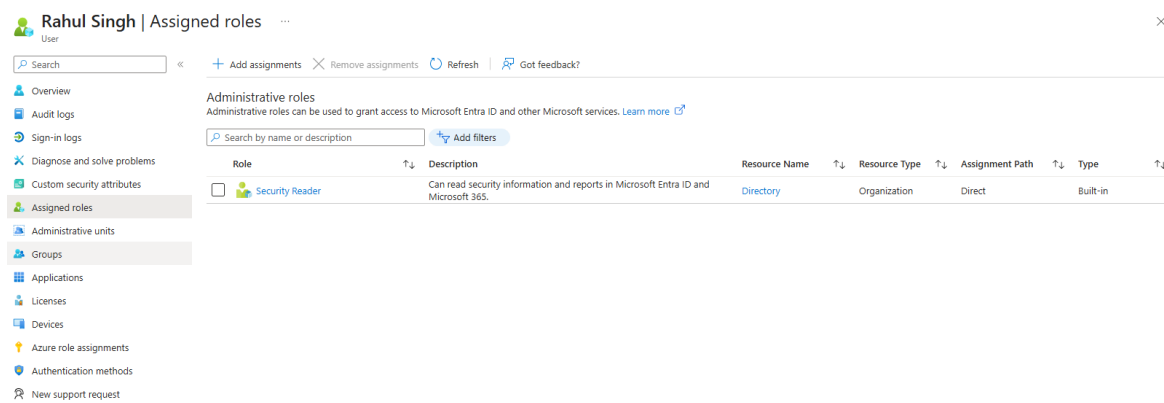


Figure 11. Rahul Singh’s Assigned Roles blade confirming the Security Reader assignment.

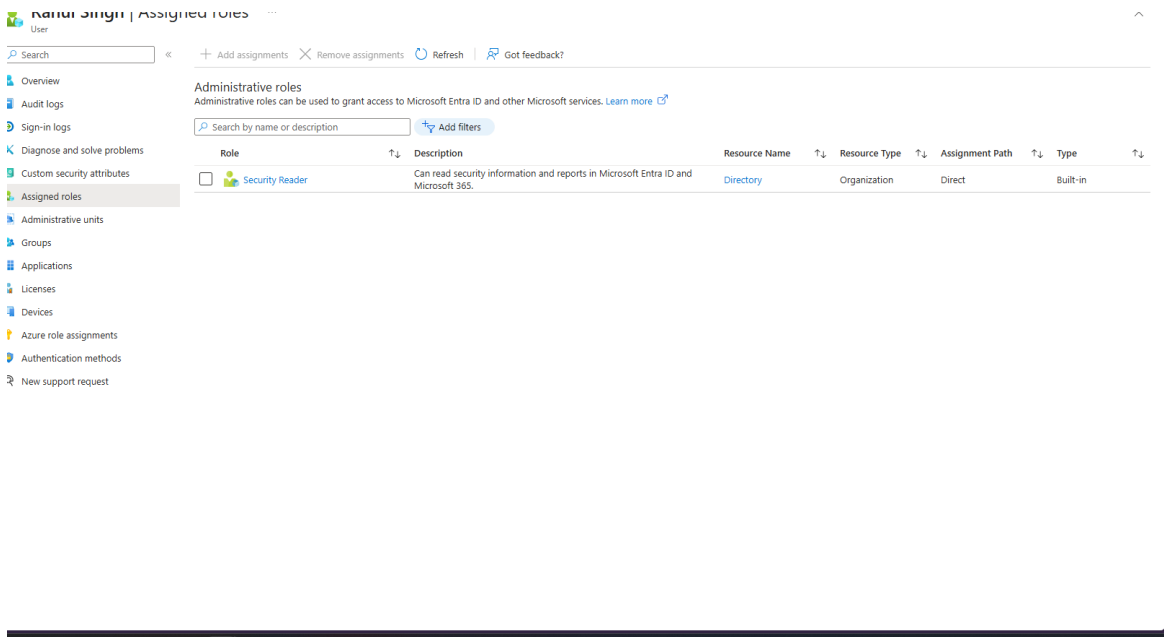


Figure 12. Assignment confirmed again from Rahul Singh’s own authenticated session.

Least Privilege in Practice

Logged in as Rahul Singh (Security Reader), the Roles and Administrators view reflects a restricted set of visible/actionable roles compared with the Global Administrator view above — demonstrating that the account holds only the permissions required for security monitoring.

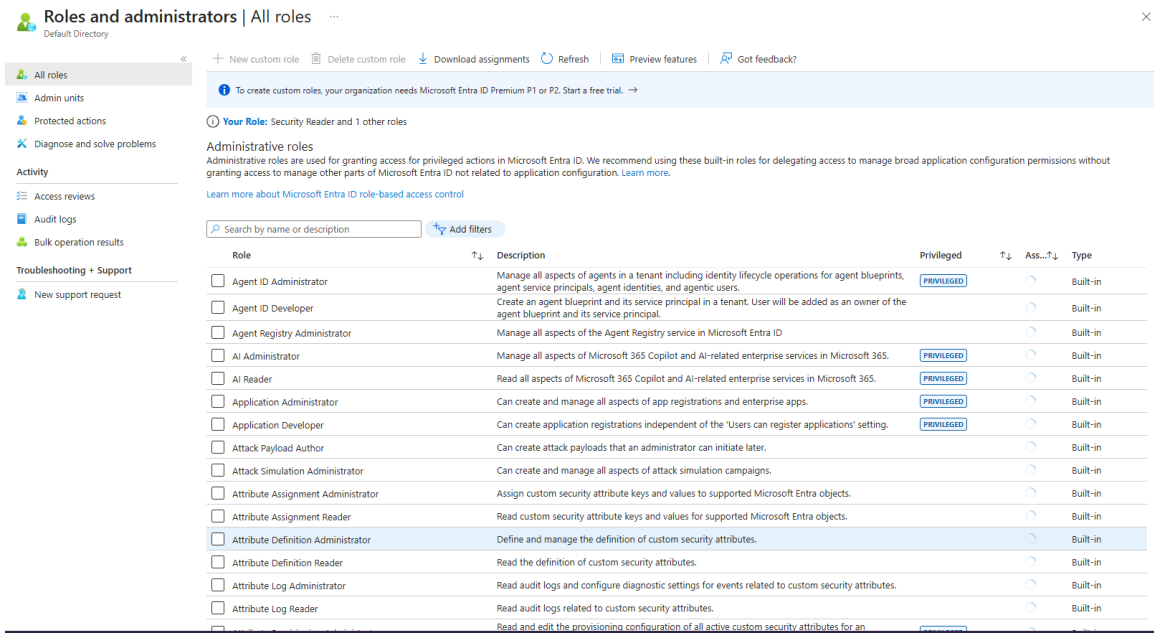


Figure 13. Roles and Administrators view as seen from the Security Reader account.

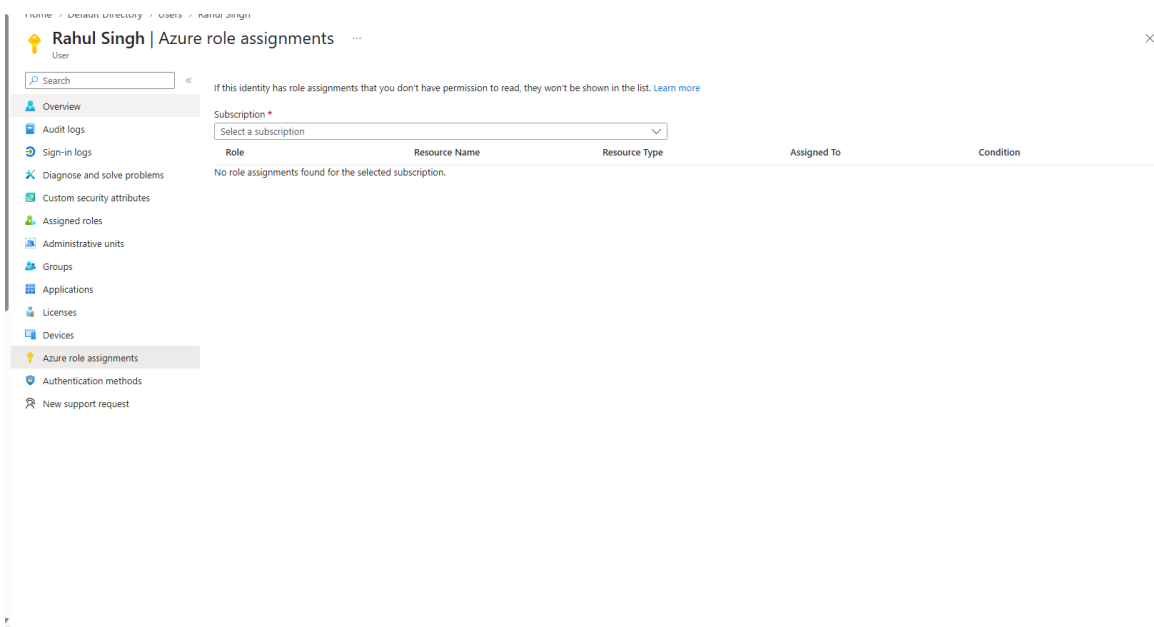


Figure 14. No Azure resource-level role assignments exist for Rahul Singh — access is scoped to directory security data only.

Access Lifecycle Management

The Security Reader assignment was later removed to demonstrate that privileged/elevated access should not be left standing once it is no longer required.

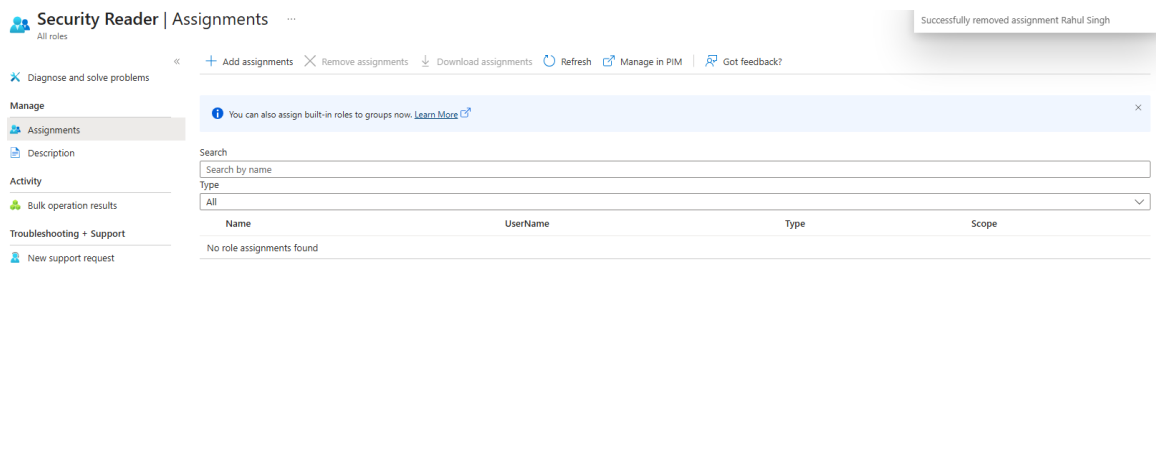


Figure 15. Security Reader assignment successfully removed from Rahul Singh.

Security Principle Demonstrated

Least privilege: give the analyst only the permissions required for security monitoring rather than making the account a Global Administrator.

2. MFA Implementation

Microsoft Authenticator was configured as an authentication method. The lab tenant’s authentication-method policy shows Microsoft Authenticator enabled for all users, and Rahul Singh successfully registered it.

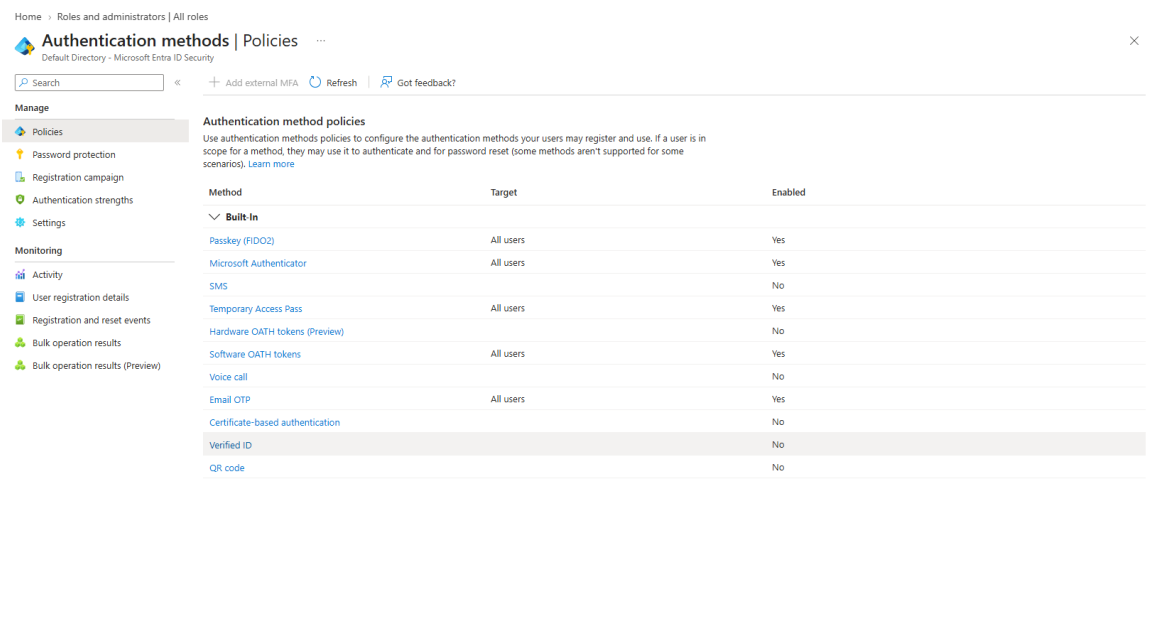


Figure 16. Tenant-wide authentication method policy — Microsoft Authenticator, Passkey (FIDO2), and Email OTP enabled.

Security Principle Demonstrated

A password alone should not be sufficient for access to sensitive administrative or security resources.

3. Authentication Method Investigation

The user’s authentication-method page was reviewed to verify the registered MFA method, the system-preferred MFA method, and the authentication-method policy in effect.

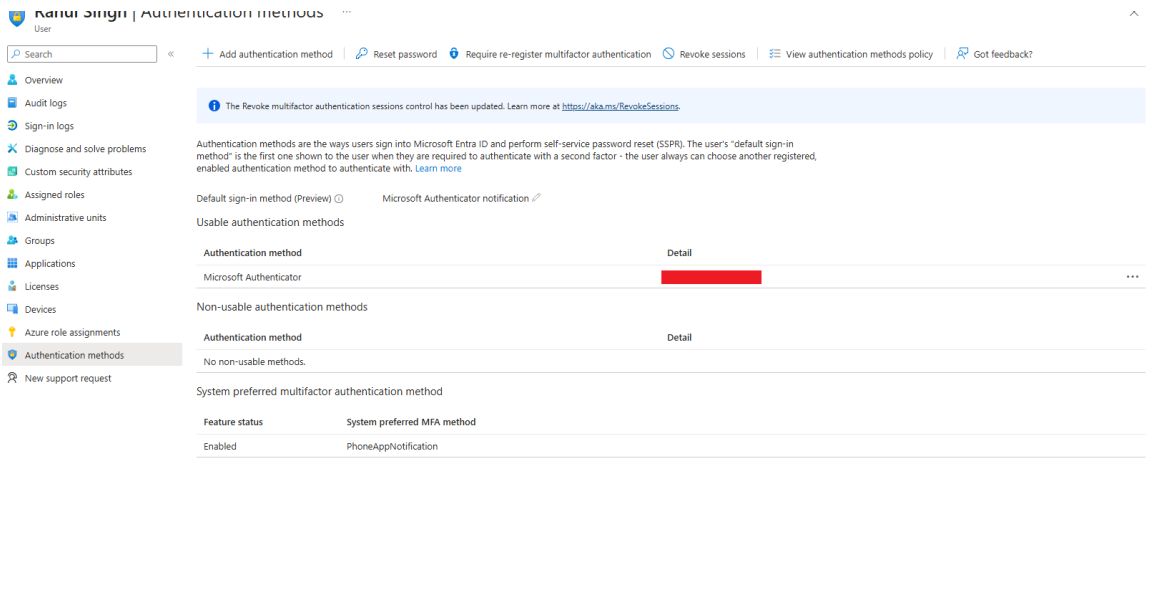


Figure 17. Rahul Singh's registered authentication methods — Microsoft Authenticator, with system-preferred MFA method set to Phone App Notification.

This demonstrates how an Entra administrator can investigate and manage a user's authentication factors.

4. Sign-In Investigation

Entra sign-in logs were used to investigate Rahul Singh's Azure Portal activity, including timestamp, application, success/interrupted status, sign-in error code, source IP/location, browser, operating system, and MFA authentication details.

Rahul Singh | Sign-in logs

Search [] Download Export Data Settings Troubleshoot Refresh Columns Got feedback?

This view will soon be replaced with a view that includes more filters infinite scrolling, and column reordering. Try out our new sign-ins preview. →

Date: **Last 24 hours** Show dates as: **Local** User contains **70c38c66-a5ff-474b-9182-ae479d7fe83b** Add filters

User sign-ins (interactive) User sign-ins (non-interactive)

Date	Request ID	User	Application	Status	Sign-in error co...	IP address	Location	Conditional Acc...	Agent T...
26/08/2026, 00:15:27	86e37308-f845-4555...	Rahul Singh	Azure Portal	Success	0			Not Applied	Not Age
26/08/2026, 00:11:26	f7299d40-0453-4eb...	Rahul Singh	Azure Portal	Success	0			Not Applied	Not Age
26/08/2026, 00:11:18	88097304-962b-478...	Rahul Singh	Azure Portal	Interrupted	50140			Not Applied	Not Age
26/08/2026, 00:09:58	30f8c694-f822-444e...	Rahul Singh	Azure Portal	Interrupted	50055			Not Applied	Not Age

Figure 18. Sign-in logs for Rahul Singh showing successful and interrupted Azure Portal sign-ins (IP address and location redacted).

Activity Details: Sign-ins

Basic info Location Device info **Authentication Details** Conditional Access Report-only

Authentication Policies Applied

App requires multifactor authentication

Date	Authentication met...	Authentication met...	Succeeded	Result detail	Requirem...
26/08/2026, 00:15:27	Previously satisfied		true	MFA requirement satis...	

Figure 19. Sign-in activity detail — Authentication Details tab confirming the MFA requirement was satisfied.

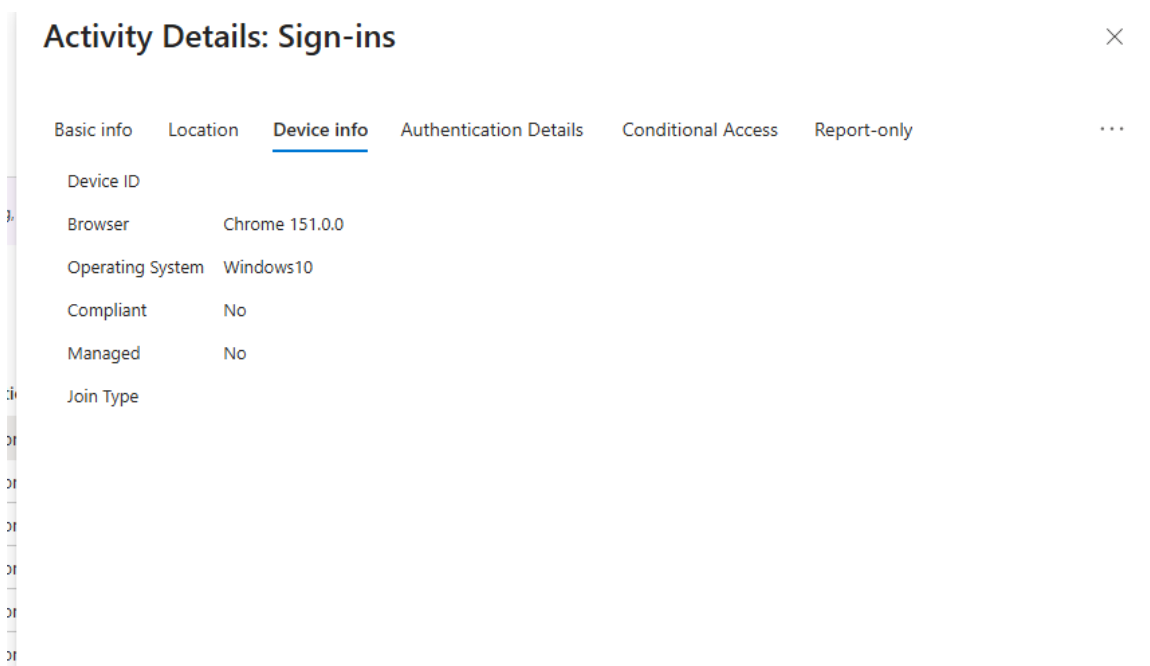


Figure 20. Sign-in activity detail — Device Info tab showing browser and operating system.

Investigation Workflow

```

User sign-in
↓
Check application
↓
Check success/failure
↓
Check source IP and location
↓
Check device/browser
↓
Check MFA result
↓
Determine whether activity appears expected
  
```

This is representative of an identity-security / SOC investigation workflow.

5. Audit Log Investigation

Microsoft Entra audit logs were reviewed for identity-management events, including add user, add member to group, add member to role, password changes, authentication-method registration, and MFA/security-information changes.

Rahul Singh | Audit logs

Search [] Download Refresh Columns Got feedback?

This view will soon be replaced with a view that includes custom security attribute logs, infinite scrolling, and column reordering. Try out our new audits preview. →

Date: Last 7 days Show dates as: Local Service: All Category: All Activity: All Add filters

Date	Service	Category	Activity	Status	Status reason	Target(s)	Initiated by (actor)
26/08/2026, 00:15:32	B2C	Authentication	Validate user authenticati...	Success	Token is valid	00000000-0000-0000-00...	security.analyst01@mailh...
26/08/2026, 00:15:20	Authentication Methods	UserManagement	User registered security L...	Success	User registered security L...	Rahul Singh	Rahul Singh
26/08/2026, 00:15:20	Core Directory	UserManagement	Update user	Success		security.analyst01@mailh...	Azure Credential Configu...
26/08/2026, 00:15:20	Authentication Methods	UserManagement	User registered all requir...	Success	User registered all requir...	Rahul Singh	Rahul Singh
26/08/2026, 00:15:15	Authentication Methods	UserManagement	User registered security L...	Success	User registered Authentic...	Rahul Singh	Rahul Singh
26/08/2026, 00:15:13	Authentication Methods	UserManagement	User registered security L...	Success	User registered security L...	Rahul Singh	Rahul Singh
26/08/2026, 00:15:13	Core Directory	UserManagement	Update user	Success		security.analyst01@mailh...	Azure Credential Configu...
26/08/2026, 00:15:02	Authentication Methods	UserManagement	User started security info ...	Success	User started the registrati...	Rahul Singh	Rahul Singh
26/08/2026, 00:14:48	Core Directory	UserManagement	Update user	Success		security.analyst01@mailh...	Azure MFA StrongAuthen...
26/08/2026, 00:14:47	Core Directory	UserManagement	Update user	Success		security.analyst01@mailh...	Azure MFA StrongAuthen...
26/08/2026, 00:11:17	Self-service Password Ma...	UserManagement	Change password (self-se...	Success	None	Rahul Singh	security.analyst01@mailh...
26/08/2026, 00:11:17	Core Directory	UserManagement	Update StsRefreshTokenV...	Success		security.analyst01@mailh...	Microsoft password reset...
26/08/2026, 00:11:17	Core Directory	UserManagement	Change user password	Success		security.analyst01@mailh...	Microsoft password reset...
26/08/2026, 00:11:17	Core Directory	UserManagement	Update PasswordProfile	Success		security.analyst01@mailh...	Microsoft password reset...
26/08/2026, 00:05:59	Core Directory	RoleManagement	Add member to role	Success		security.analyst01@mailh...	malhotra.ishaan04_gmail...
26/08/2026, 00:00:33	Core Directory	GroupManagement	Add member to group	Success		security.analyst01@mailh...	malhotra.ishaan04_gmail...
25/08/2026, 23:58:52	Core Directory	UserManagement	Add user	Success		security.analyst01@mailh...	malhotra.ishaan04_gmail...
25/08/2026, 23:58:52	Core Directory	UserManagement	Update PasswordProfile	Success		security.analyst01@mailh...	malhotra.ishaan04_gmail...

Figure 21. Audit log for Rahul Singh showing role, group, and authentication-method management events, each with initiating actor and status.

The logs were used to establish who performed an action, what changed, and whether the operation succeeded.

6. PAM / PIM Evaluation

Microsoft Entra Privileged Identity Management (PIM) was investigated as the PAM component of the lab. The PIM interface was successfully accessed and the Microsoft Entra roles section was reviewed.

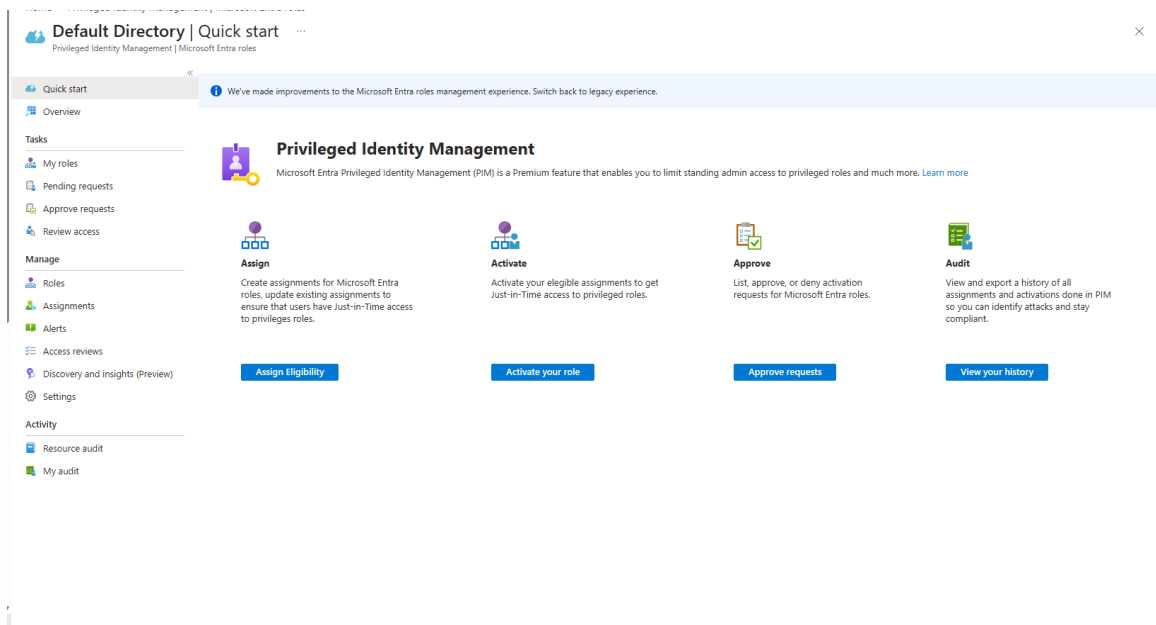
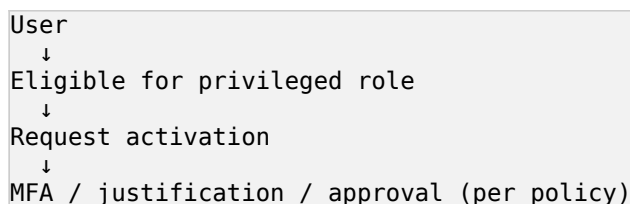


Figure 22. Privileged Identity Management Quick Start — Assign, Activate, Approve, and Audit tasks.

Role	Description	Active	Eligible
Agent ID Administrator	Manage all aspects of agents in a tenant including identity lifecycle operations for agent blueprints, agent...	-	-
Agent ID Developer	Create an agent identity blueprint and its agent identity blueprint principal in a tenant. User will be added...	-	-
Agent Registry Administrator	Manage all aspects of the Agent Registry service in Microsoft Entra ID	-	-
AI Administrator	Manage all aspects of Microsoft 365 Copilot and AI-related enterprise services in Microsoft 365.	-	-
AI Reader	Read all aspects of Microsoft 365 Copilot and AI-related enterprise services in Microsoft 365.	-	-
Application Administrator	Can create and manage all aspects of app registrations and enterprise apps.	-	-
Application Developer	Can create application registrations independent of the 'Users can register applications' setting.	-	-
Attack Payload Author	Can create attack payloads that an administrator can initiate later.	-	-
Attack Simulation Administrator	Can create and manage all aspects of attack simulation campaigns.	-	-
Attribute Assignment Administrator	Assign custom security attribute keys and values to supported Microsoft Entra objects.	-	-
Attribute Assignment Reader	Read custom security attribute keys and values for supported Microsoft Entra objects.	-	-
Attribute Definition Administrator	Define and manage the definition of custom security attributes.	-	-
Attribute Definition Reader	Read the definition of custom security attributes.	-	-
Attribute Log Administrator	Read audit logs and configure diagnostic settings for events related to custom security attributes.	-	-
Attribute Log Reader	Read audit logs related to custom security attributes.	-	-
Attribute Provisioning Administrator	Read and edit the provisioning configuration of all active custom security attributes for an application.	-	-
Attribute Provisioning Reader	Read the provisioning configuration of all active custom security attributes for an application.	-	-
Authentication Administrator	Can access to view, set and reset authentication method information for any non-admin user.	-	-
Authentication Extensibility Administrator	Customize sign in and sign up experiences for users by creating and managing custom authentication ext...	-	-
Authentication Extensibility Password Administrator	Trigger a password submit or password update event for custom authentication.	-	-
Authentication Policy Administrator	Can create and manage the authentication methods policy, tenant-wide MFA settings, password protection...	-	-
Azure AD joined Device Local Administrator	Users assigned to this role are added to the local administrators group on Microsoft Entra joined devices.	-	-

Figure 23. PIM Microsoft Entra roles view listing eligible/active assignment columns for all built-in roles.

Intended PIM Workflow



↓
Temporary privileged access
↓
Access expires
↓
Audit trail

This provides a stronger security model than permanent, standing privileged access.

Licensing Limitation

The lab tenant uses Microsoft Entra ID Free. When attempting to create a PIM role assignment, Microsoft Entra displayed the following message:

The tenant needs to have Microsoft Entra ID P2 or Microsoft Entra ID Governance license.

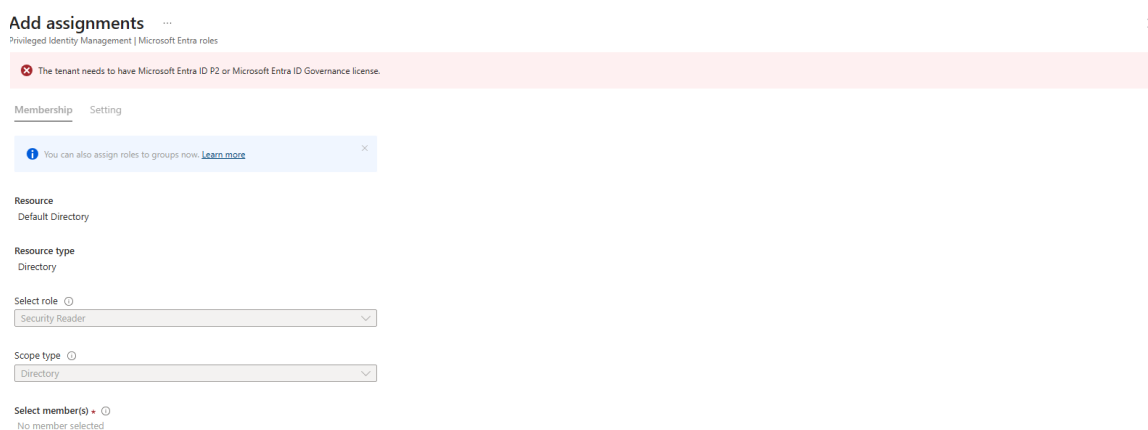


Figure 24. PIM “Add assignments” screen showing the Entra ID P2 / Governance licensing requirement.

Therefore, an actual PIM eligible/JIT assignment could not be created in this lab environment. This limitation is intentionally documented rather than presenting an unimplemented feature as completed.

What Was Successfully Demonstrated

- PIM interface access
- Microsoft Entra role management
- PIM role/assignment workflow exploration
- Understanding of eligible vs. active access
- Understanding of Just-In-Time privileged access
- Identification of the licensing requirement

What Was Not Claimed

- Actual PIM eligible assignment
- Actual JIT activation
- Actual PIM approval workflow

7. IAM vs. PAM

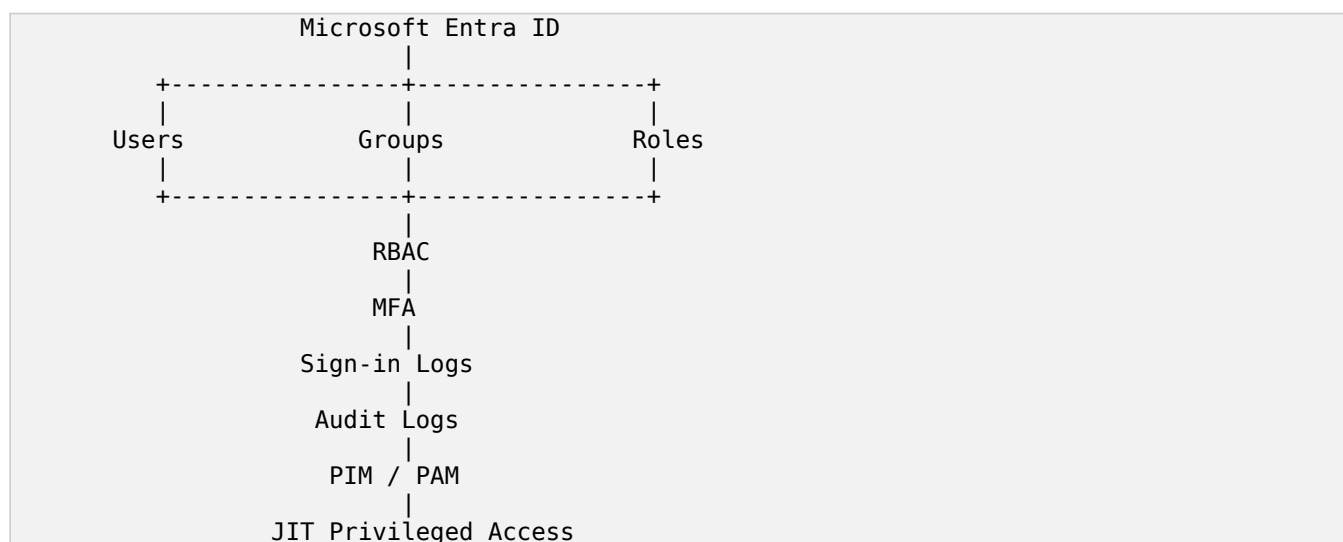
IAM	PAM / PIM
Manages identities and access	Protects privileged access
User/group management	Privileged-role management
RBAC	RBAC + JIT controls
MFA	MFA can be used during privileged activation
Least privilege	Time-limited privileged access
Access assignment	Eligibility → activation → expiration
General identity security	Higher-risk administrative access

Simple Explanation

IAM asks: Who is this user and what can they access?

PAM asks: Does this user really need privileged access right now, and how long should they have it?

8. Security Architecture



9. Skills Demonstrated

Identity Security

- Microsoft Entra ID
- User lifecycle management
- Group management
- RBAC
- Least privilege
- Authentication methods
- MFA

Security Operations

- Sign-in log analysis
- Authentication investigation
- IP/location review
- Device/browser investigation
- Audit-log analysis
- Access lifecycle review

Privileged Access Management

- PIM concepts
- Eligible vs. active assignments
- Just-In-Time access
- Temporary privileged access
- Privileged-role auditing
- Licensing/feature assessment

10. Key Security Takeaways

1. Avoid giving users Global Administrator when a narrower role is sufficient.
2. Use MFA to protect identity access.
3. Review authentication methods when investigating account activity.
4. Use sign-in logs to investigate suspicious authentication.
5. Use audit logs to establish what administrative changes occurred.
6. Remove unnecessary standing privileges.
7. Use PIM/JIT for privileged access when the required licensing is available.
8. Keep evidence honest: distinguish between features actually implemented and features evaluated but unavailable in the lab.

Project Status

Workstream	Status
IAM	Completed
RBAC	Completed
MFA	Completed
Authentication investigation	Completed
Sign-in investigation	Completed
Audit-log investigation	Completed
PAM/PIM evaluation	Completed
PIM JIT implementation	Not available in current Entra Free tenant due to licensing

Privacy Note

Tenant ID and sign-in source IP/location have been redacted in the figures above. Any remaining personal identifiers (e.g., email addresses) belong to lab test accounts created solely for this exercise.